INTRODUCTION

When Aristotle wrote about plant and animal species existing for the sake of others, he showed a basic understanding of food chains—as have countless observers of the natural world since ancient Greek times. Arab scholar al-Jahiz described a three-level food chain in the 9th century, as did the Dutch microscopist Antonie van Leeuwenhoek in 1717. British naturalist Richard Bradley published more detailed findings on food chains in 1718, and in 1859, Charles Darwin described a "web of complex relations" in the natural environment in his book *On the Origin of Species*. The concept of a food web, with many predator–prey interactions, was then further developed by Charles Elton in his classic *Animal Ecology* (1927).

The concept of the ecosystem ("a recognizable self-contained entity") followed soon after, when in 1935, British botanist Arthur Tansley wrote that organisms and their environment should be considered one physical system. In his Ph.D. thesis, American ecologist Raymond Lindeman expanded on Tansley's work, positing that ecosystems are composed of physical, chemical, and biological processes "active within a space–time unit of any magnitude."

Lindeman also conceived the idea of feeding levels, or trophic levels, each of which is dependent on the preceding one for its survival. In 1960, the American team of Nelson Hairston, Frederick Smith, and Lawrence Slobodkin published findings on the factors controlling animals on different trophic levels. They identified both the top-down pressures exerted by predators and the bottom-up pressures exerted by limitations on food supply. Twenty years later, American ecologist Robert Paine wrote of the trophic cascade effect—the way a system is changed by the removal of a key species. He described changes to the food web after the experimental removal of the ocher starfish from an intertidal zone. This predatory